

**SYNTHETIC COURT DOCUMENT — FICTIONAL CASE — NOT AN
ACTUAL FILING**

UNITED STATES DISTRICT COURT FOR THE WESTERN DISTRICT OF VIRGINIA

Civil Action No. SYN-WDVA-25-CV-0733

Professor Adrian Crane Video and Human-Factors Expert Report and Deposition Excerpts

Report dated October 30, 2025; designated with Ellison's December 12, 2025 opposition.

Scope, materials, and method

I was retained by Mara Ellison to examine the integrity and perceptual limits of police video and audio from February 3, 2025. My work concerns media authentication, camera geometry, temporal alignment, visual occlusion, and cautious interpretation under rapid-event conditions. I do not decide whether force was lawful or which witness is credible.

I reviewed the native exports produced by Sergeant Lila Grant, export manifests and hashes, device metadata, Rusk and Vega recordings, dispatch audio, conducted-energy-weapon data, photographs of the breezeway, reports, and relevant deposition excerpts. I used ordinary frame inspection, waveform comparison, and scene measurements. I did not use generative fill, pose estimation, optical-flow reconstruction, or artificial-intelligence inference to fabricate views that the cameras never captured.

The files are usable for timing and for visible or audible events within their limits. Their authenticity does not imply completeness. A genuine recording can leave a decisive action hidden.

Crane report — source integrity and clock alignment

Grant's export records preserve the original item identifiers, creation data, export time, and cryptographic hashes. The supplied media match those manifests. I found no cut, missing encoded segment, reordered frame group, or unexplained discontinuity in either officer's file. That supports integrity of the copies reviewed.

I aligned the two camera streams using repeated acoustic transients and visible events, then compared them to dispatch and device clocks. Ordinary clock offsets were documented rather than silently erased. On the common event timeline, Rusk's arrival is approximately 19:52:14, initial contact occurs near 19:52:48, wrist control begins around 19:53:02, and the most restricted view extends from 19:53:08 until about 19:53:22.

The conducted-energy device records one five-second drive-stun cycle from approximately 19:53:16 through 19:53:21. The image clears just after the cycle. Those temporal relationships are supported even though the initiating hand movements are not visible. Timing can establish that events overlap without revealing their unseen physical content. A synchronized gap remains a gap.

Crane report — Rusk camera field of view

Rusk's chest-mounted camera provides a useful wide view during approach. At contact distance, however, its lens position is not equivalent to his eyes. His arms, clothing, Ellison's body, and rotation of the torso successively block the relevant lower-center region. Motion blur and rapid exposure changes further reduce fine detail.

Beginning around 19:53:08, the recording does not continuously display Ellison's hands, Rusk's forearm, and the duty-belt area in a single interpretable view. Some frames contain fragments of an arm or fabric. A fragment cannot reliably identify whose limb it is, its depth relative to the belt, or whether contact occurred outside the plane captured by the lens.

The obstruction is physical, not evidence of file alteration. Increasing brightness or contrast can make recorded pixels easier to see but cannot reveal an object covered by another object. Slowing playback likewise lengthens viewing time without adding spatial information. I therefore cannot infer from Rusk's file whether Ellison struck his forearm, reached for equipment, or only pulled and recoiled from wrist pressure.

Crane report — Vega camera geometry

Vega's camera supplies a different angle, but not a complete corrective angle. She approached from the stair side of a narrow breezeway. The railing, Rusk's shoulder and back, and Ellison's changing height create overlapping occlusions. Her lens intermittently captures upper-body orientation while the hand-and-belt area remains screened.

Triangulation requires simultaneous views of the same feature with adequate resolution and known geometry. These recordings do not meet those conditions for the disputed hands. A glimpse in one stream cannot be joined to an unrelated fragment in the other to construct a continuous motion path. The cameras also have rolling perspective and wearer movement rather than fixed calibrated positions.

Vega's testimony that she lacked a reliable view of the hands is consistent with the recorded geometry. It is neither proof that no strike or reach occurred nor corroboration that one did. The second camera narrows some timing and position questions, but it does not eliminate the central uncertainty. Describing it as a “second angle” without stating these limits would overstate the record.

Crane report — audio content and attribution limits

Both officer files record commands, Ellison's cry, speech overlap, contact noise, and changes in breathing. Around 19:53:11, “stop pulling” and “you're hurting me” occur close together. The audio supports that Rusk perceived or described pulling and that Ellison expressed pain. It does not encode a visible description of her hands.

Short impact-like sounds may arise from shoes, equipment, clothing, the railing, bodies, or the camera housing. Without a visible source or a distinctive calibrated signature, I cannot label a particular transient as a forearm strike. Nor can directional localization from two moving body microphones reliably place a hand at the duty belt.

Listeners are vulnerable to expectation effects: a caption or report can make an ambiguous noise seem to confirm the described event. My transcript identifies audible words and uncertainty but does not caption disputed physical acts. The device activation has a recognizable temporal marker, and the cry during the cycle can be placed in sequence. That does not retrospectively disclose why Rusk chose to activate it.

Crane report — human perception in compressed encounters

An officer may perceive an event that a chest camera misses because eyes and lens occupy different locations. Conversely, a camera may preserve detail an officer did not attend to. Video is not a transparent copy of perception, and a witness's asserted perception is not proven merely because the recording fails to contradict it.

Rapid motion, narrowed attention, stress, close distance, and visual crowding can affect encoding and later memory. Those principles caution against assuming perfect recollection by any witness. They do not authorize me to declare that Rusk's account is false or that Ellison's account is accurate. Each account must be compared with evidence that actually bears on it.

Repeated slow-motion viewing can also create hindsight confidence. A reviewer knows an activation is coming and can search ambiguous frames for a precursor. That task differs from perception in real time. My demonstrative timeline uses neutral event descriptions: wrist grasp, view restricted, audible statements, device cycle, image clears, and handcuffing. It does not insert “strike” or “belt reach” where the image lacks sufficient information.

Crane report — rejected reconstruction methods

Counsel asked whether modern enhancement could settle the dispute. Resolution enhancement may resample or sharpen visible edges, but it cannot recover a hand hidden behind an opaque body. Generative systems can produce plausible pixels based on learned patterns; plausible is not observed. Such output would be a synthetic illustration, not evidence of this encounter.

Pose-estimation software similarly depends on visible landmarks and model assumptions. During the critical interval, relevant landmarks disappear, overlap, or fall below usable resolution. A model might interpolate a likely arm path, but multiple paths fit the same recorded fragments. Selecting one would embed an assumption about the very fact in controversy.

I therefore preserved native frames, documented ordinary viewing adjustments, and stopped at the epistemic boundary of the media. The defensible conclusion is not that “nothing happened” inside the obstruction. It is that these files do not reliably determine whether Ellison struck Rusk, moved toward the duty belt, or confined her movement to pulling and recoil. Any factual finding must come from the whole admissible record, not manufactured imagery.

Crane deposition excerpt — cross-examination

Q. You agree the video does not disprove Officer Rusk's account, correct?

A. Correct. It also does not prove that account. The contested actions occur where neither camera provides a dependable view of the hands and belt relationship.

Q. Isn't Ellison visibly moving rather than perfectly still?

A. Yes. The recordings show gross body movement and a close physical encounter. Ellison does not claim complete stillness; she acknowledges pulling or recoiling from the wrist grasp. Gross motion does not identify a forearm strike, deliberate equipment reach, or immediate threat.

Q. Could the phrase “stop pulling” corroborate Rusk?

A. It corroborates that he uttered those words and is compatible with some pulling. It cannot distinguish his broader description from Ellison's narrower one. Audio interpretation must not promote a speaker's contemporaneous characterization into an independently verified image.

Q. Did you alter playback speed for the demonstrative?

A. I created synchronized ordinary-speed and slowed views, clearly labeled. Slowing facilitates inspection but does not change source timing or generate missing content. The native exports and hashes remain the controlling objects.

Crane report — conclusions and use restrictions

The media are authentic copies suitable for bounded uses. They establish the general sequence, support a single drive-stun cycle, record key words and distress, and show when physical contact, takedown, and handcuffing occurred. They also document their own limitation: the central hand- and-belt region is not reliably visible during the decisive fourteen seconds.

Neither officer view conclusively resolves resistance or threat. Audio does not cure the visual deficit, and ordinary enhancement cannot see through bodies or clothing. Automated reconstruction would add model-generated assumptions and is not a valid factual source. My demonstratives must remain paired with the limitation statement and may not be relabeled as recovered footage.

I offer no opinion on force reasonableness, clearly established law, credibility, or appellate jurisdiction. A court may state assumed facts for legal analysis, but that statement does not convert an obstructed recording into factual proof. Conversely, acknowledging uncertainty does not prevent a factfinder from considering testimony and other evidence. My conclusion is precise: the recordings constrain chronology while leaving the parties' pivotal accounts genuinely unresolved.